



School: Campus:
Academic Year: Subject Name: Subject Code:
Semester: Program: Branch: Specialization:
Date:

Applied and Action Learning

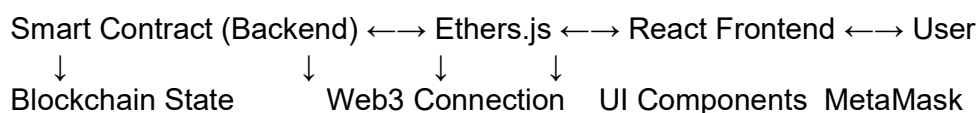
(Learning by Doing and Discovery)

Name of the Experiment :

* Coding Phase: Pseudo Code / Flow Chart / Algorithm

UI for DApps – Building a DApp Frontend :

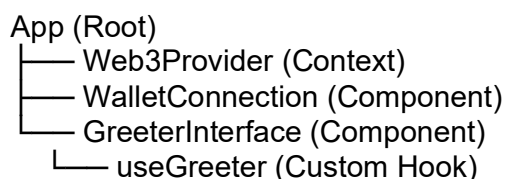
Overall Architecture Strategy:



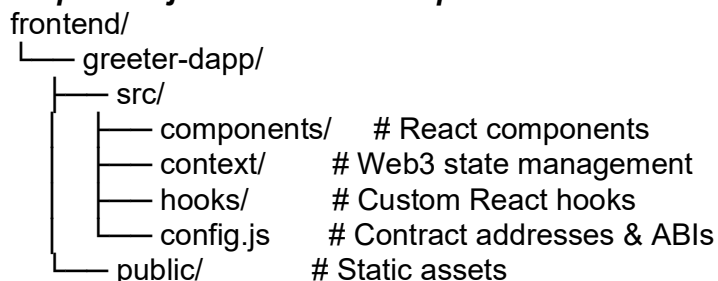
Technical Stack Selection :

1. Frontend Framework: React
 - Why: Most popular, extensive ecosystem, great for DApps
 - Alternative: Vue.js, Svelte, or vanilla JS
2. Web3 Library: Ethers.js
 - Why: Simpler API than web3.js, better TypeScript support
 - Key Features: Contract interaction, wallet connection, event handling
3. UI Library: Material-UI (MUI)
 - Why: Professional design, responsive, accessible components
 - Alternative: Chakra UI, Ant Design, or custom CSS
4. State Management: React Context + Hooks
 - Why: Built-in React solution, perfect for Web3 state
 - Alternative: Redux, Zustand, or Recoil

Component Architecture Design :



Step 1: Project Structure Setup



Coding Phase: Pseudo Code / Flow Chart / Algorithm

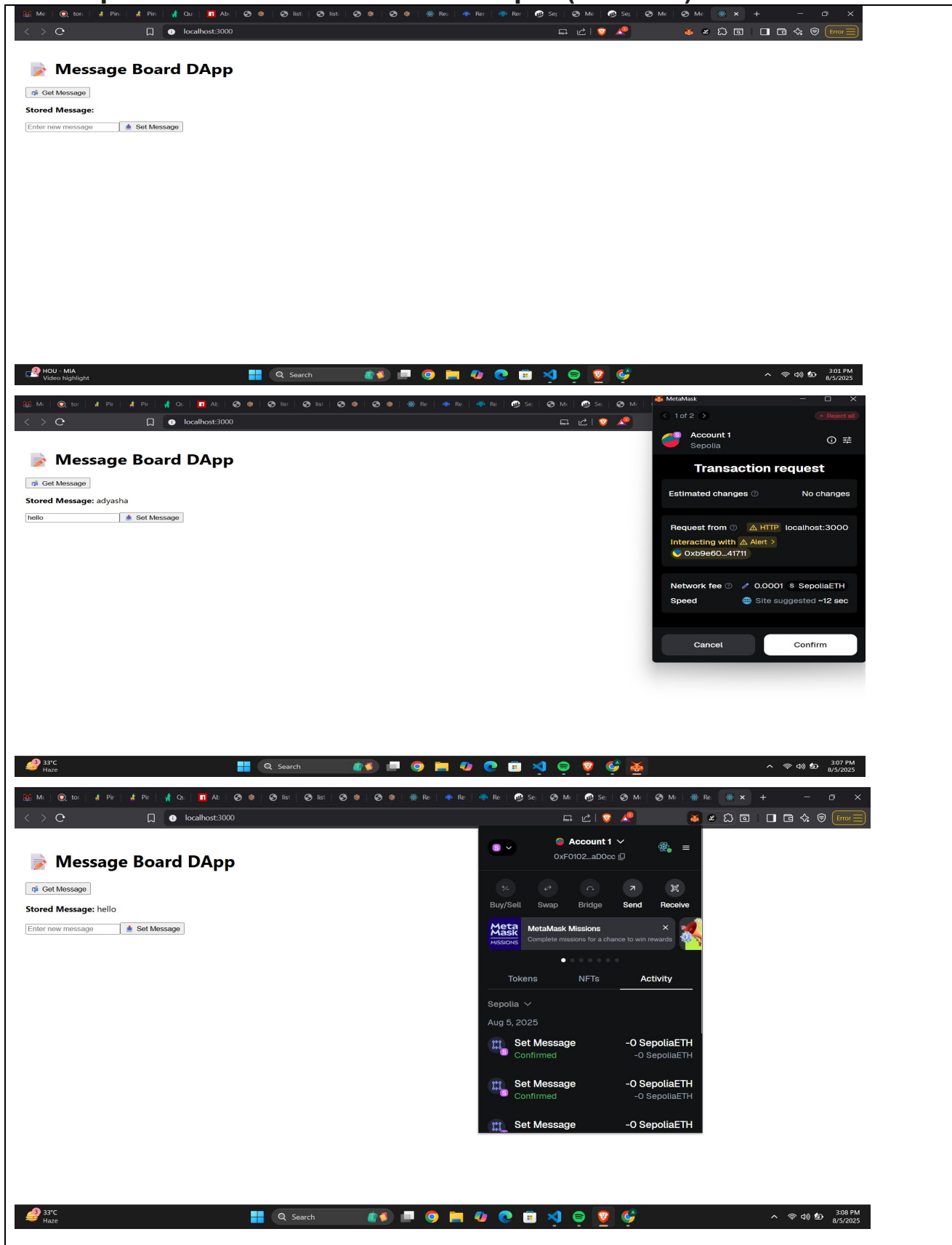
Key Design Decisions

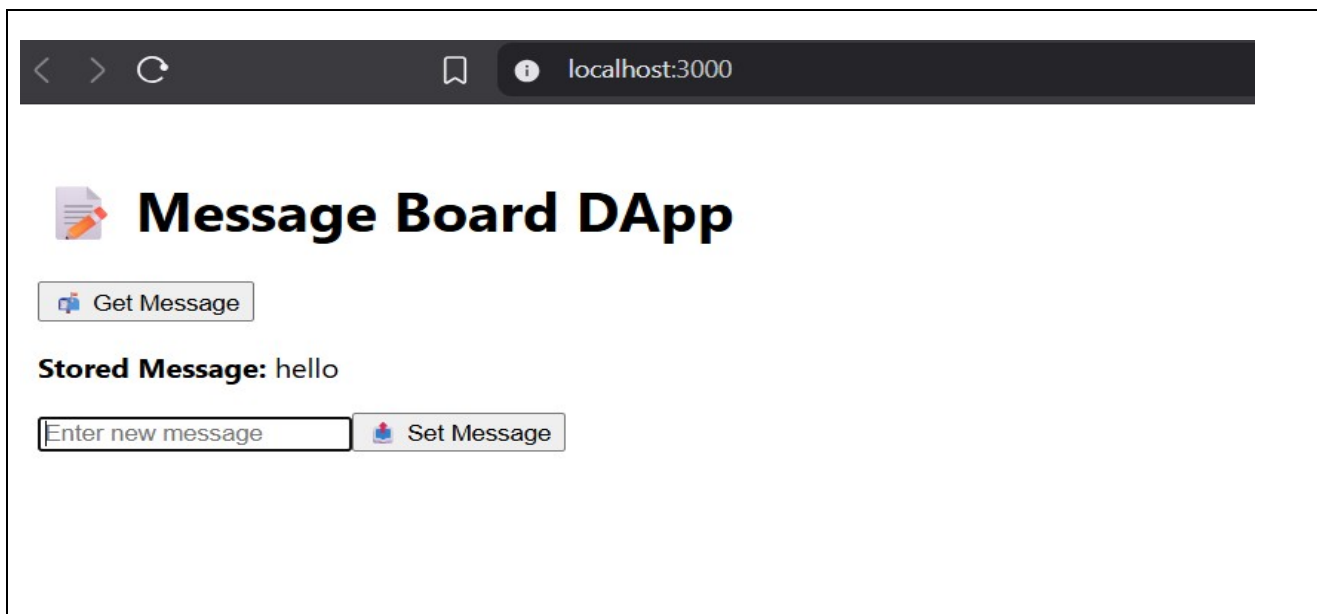
1. Responsive Design Approach
 - Used Material-UI's grid system
 - Mobile-first responsive design
 - Accessible color contrasts and typography
2. User Experience Considerations
 - Loading states for all async operations
 - Clear feedback for transactions (pending, success, error)
 - Intuitive workflow from connection to interaction
 - Error handling with helpful messages
3. Developer Experience
 - Modular architecture for easy maintenance
 - Custom hooks for reusable logic
 - Configuration centralization in one file
 - Type safety with proper prop validation

* Softwares used

- VS Code
- Solidity, Hardhat
- Javascript
- Ether.js
- Metamask

Implementation Phase: Final Output (no error)





* Observations

1. Material-UI Effectiveness:
 - Rapid development with pre-built components
 - Consistent design across application
 - Accessibility benefits out-of-the-box
 - Customization limitations for brand identity
2. Responsive Design Challenges:
 - Mobile wallet interactions different from desktop
 - Transaction details display on small screens
 - Connection status visibility across devices
 - Touch targets for critical actions

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Name :

Regn. No. :

Signature of the Faculty:

Page No.....